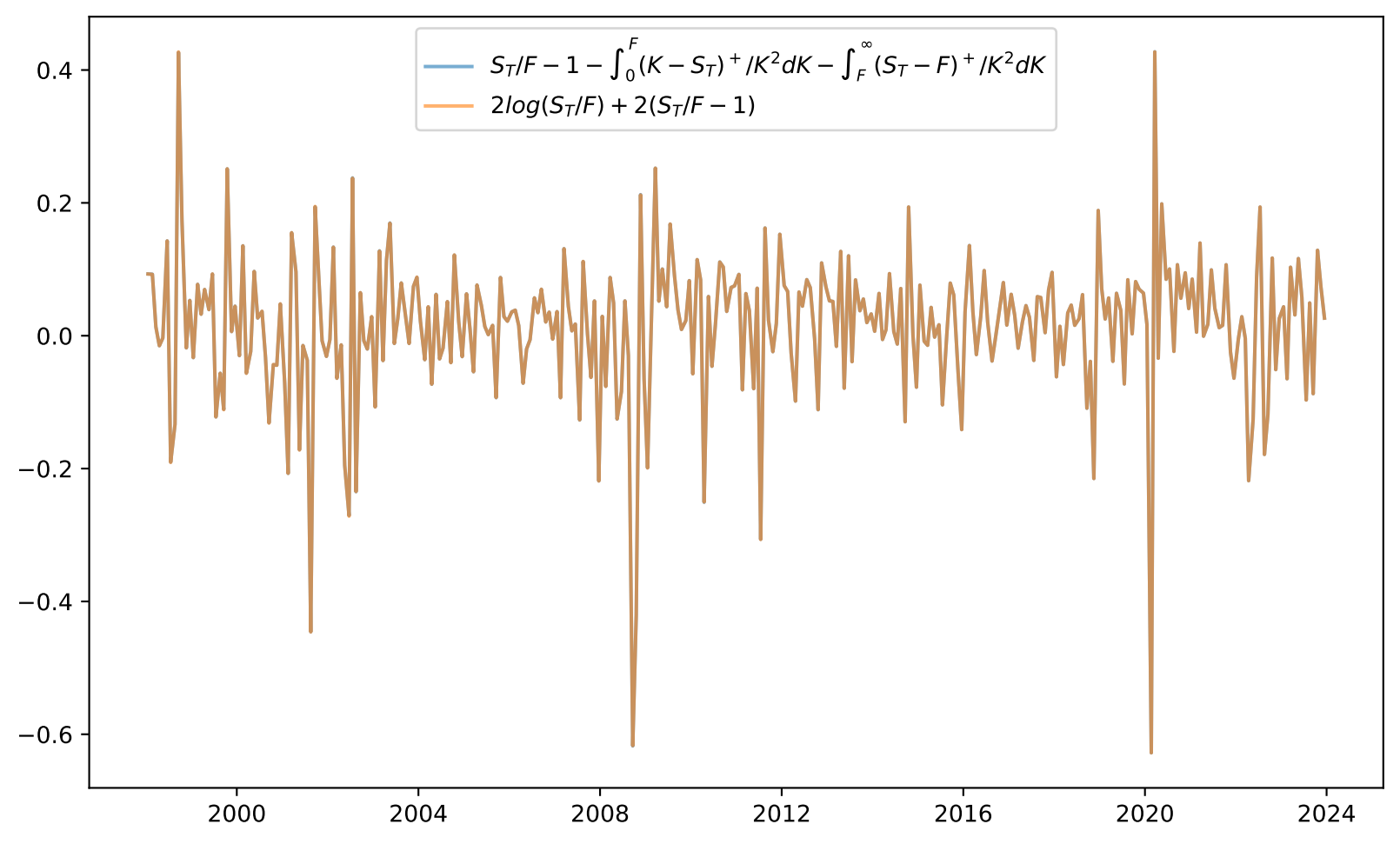
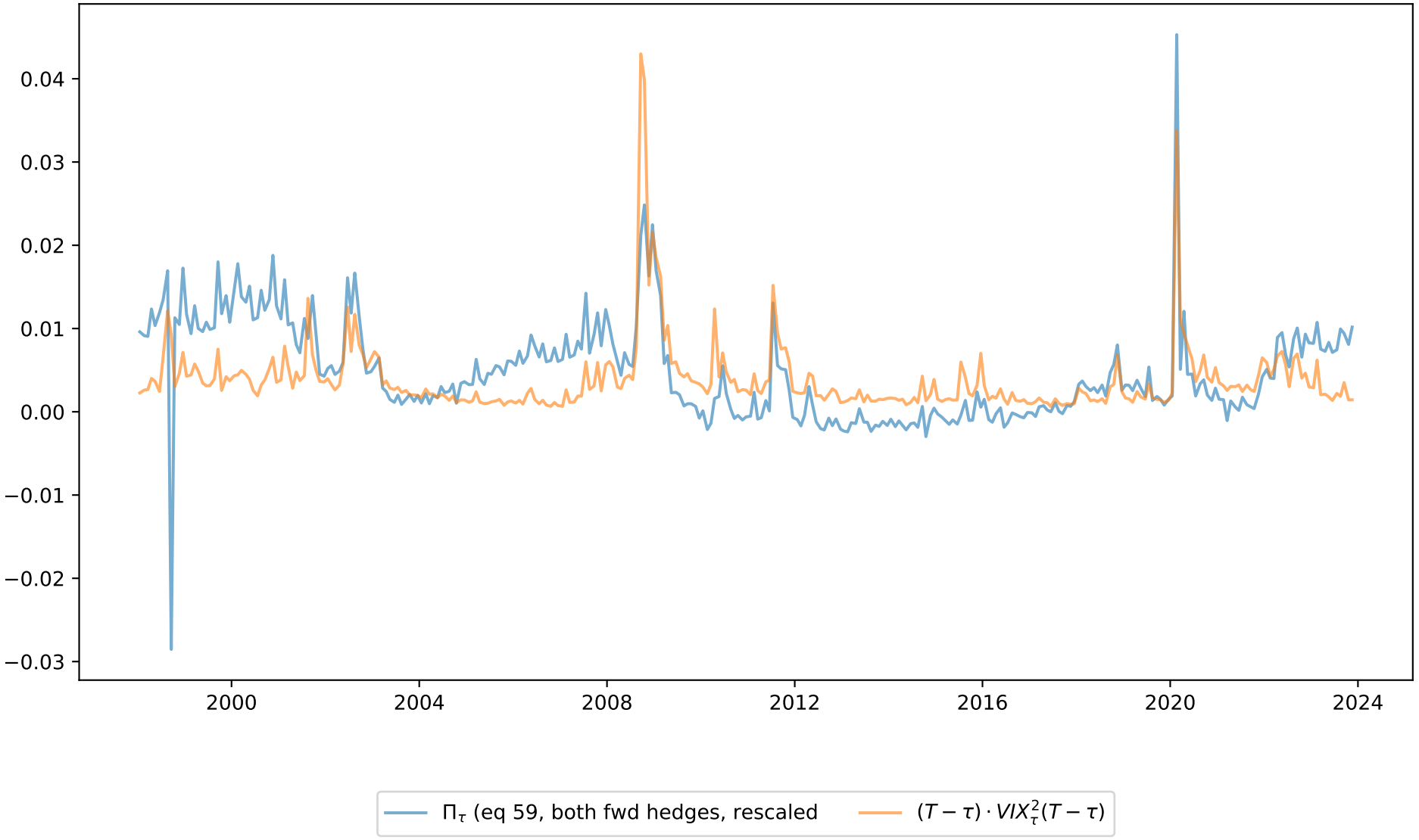

$S_T/F - 1 - \int_0^F (K - S_T)^+ / K^2 dK - \int_F^\infty (S_T - F)^+ / K^2 dK$
$2log(S_T/F) + 2(S_T/F - 1)$
0.4
0.2
0.0
−0.2
−0.4
−0.6
2000
2004
2008
2012
2016
2020
2024

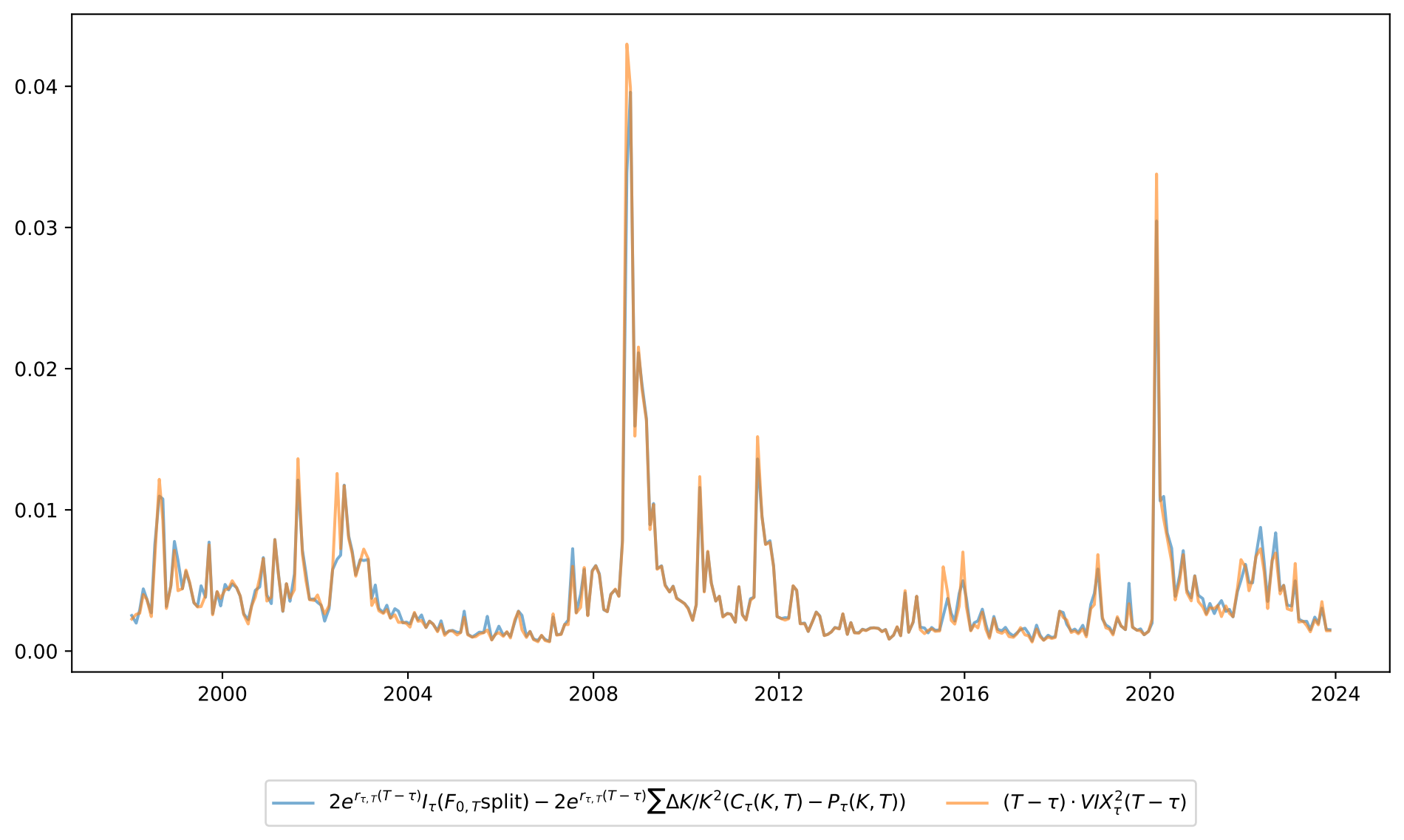
log_actual	log_approx
1.0	0.999998397847608
0.999998397847608	1.0

log_actual	log_approx
312.0	312.0
0.01416	0.01417
0.111904	0.111902
-0.627892	-0.627951
-0.029994	-0.030073
0.02718	0.027081
0.072319	0.072412
0.427279	0.427512


0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
2000
2004
2008
2012
2016
2020
2024
Π_τ (eq 59, both fwd hedges, rescaled
$(T-\tau)\cdot VIX_\tau^2(T-\tau)$

approximation_full_hedge	future_VIX^2
1.0	0.583899409863340 1
0.583899409863340 1	1.0

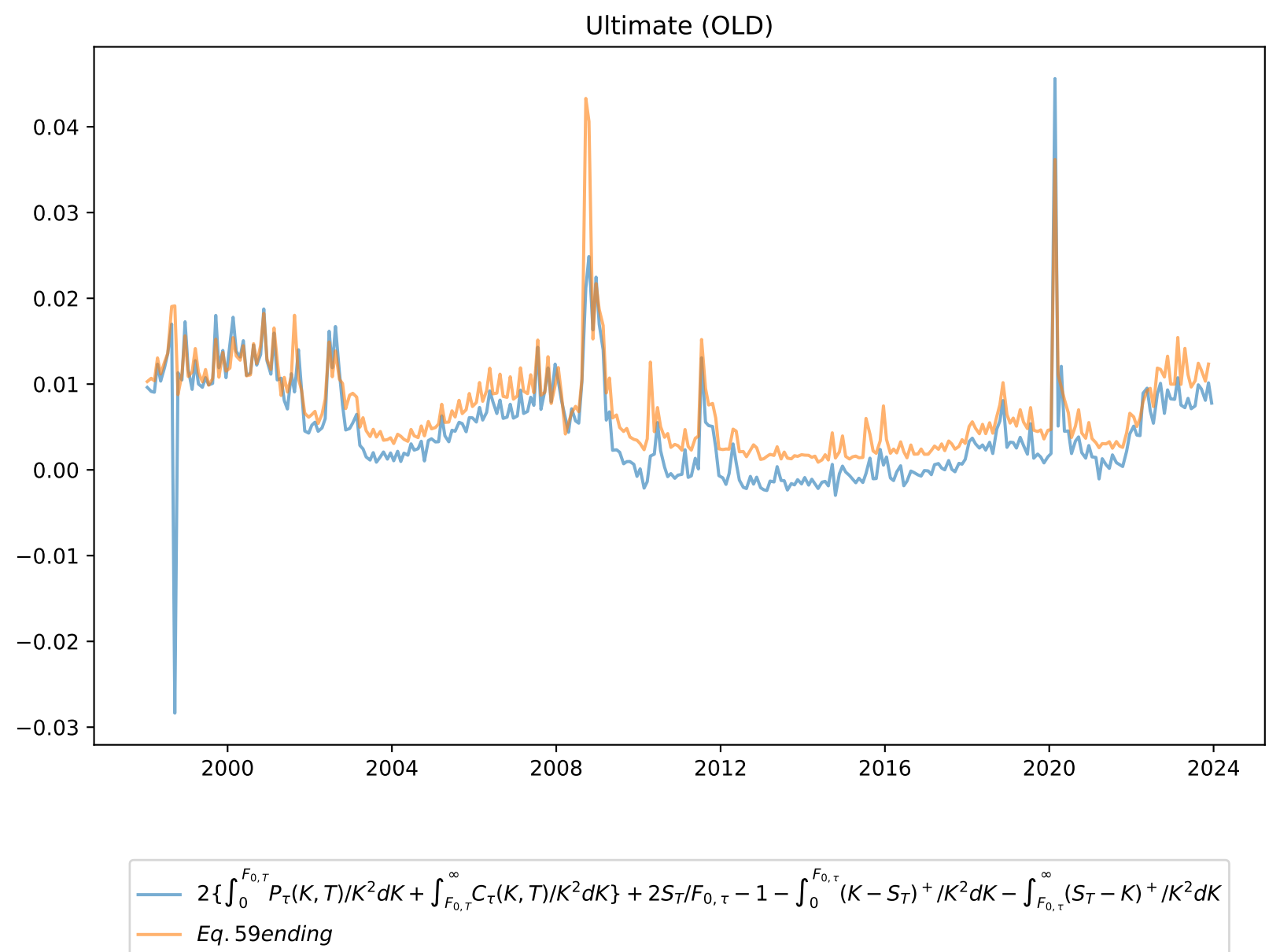
approximation_full_hedge	future_VIX^2
311.0	311.0
0.004732	0.003899
0.006075	0.004552
-0.028537	0.000638
0.000567	0.001546
0.003303	0.002676
0.008137	0.004442
0.045296	0.042988


0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2e^{r_{\tau,T}(T-\tau)}I_\tau(F_{0,T}\text{split}) - 2e^{r_{\tau,T}(T-\tau)}\sum \Delta K/K^2(C_\tau(K,T) - P_\tau(K,T))$
$(T-\tau)\cdot VIX_\tau^2(T-\tau)$

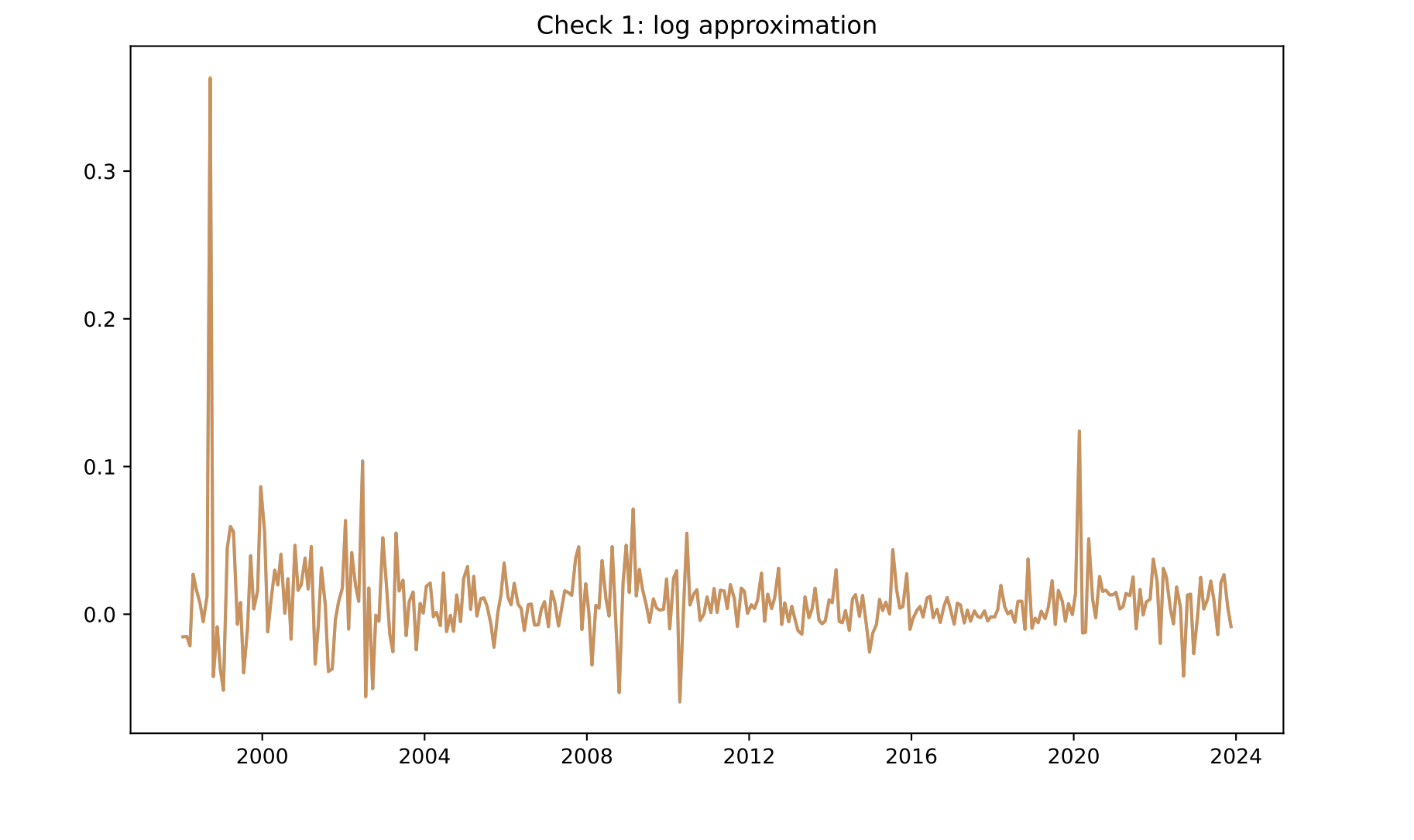
approximation_substitution	future_VIX^2
1.0	0.986221280895758
0.986221280895758	1.0

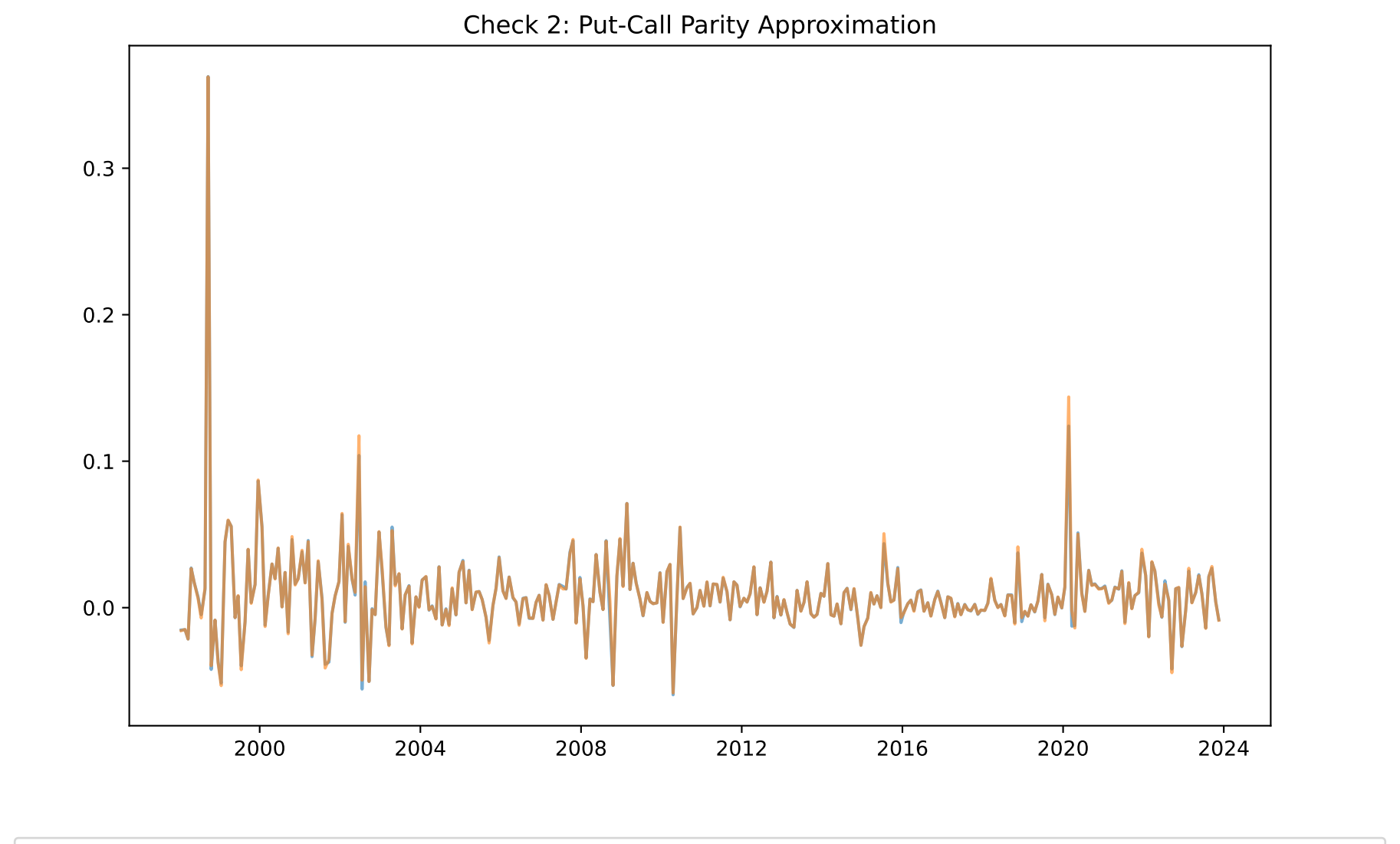
approximation_substitution	future_VIX^2
311.0	311.0
0.003927	0.003899
0.004196	0.004552
0.000697	0.000638
0.001671	0.001546
0.002799	0.002676
0.004613	0.004442
0.039583	0.042988

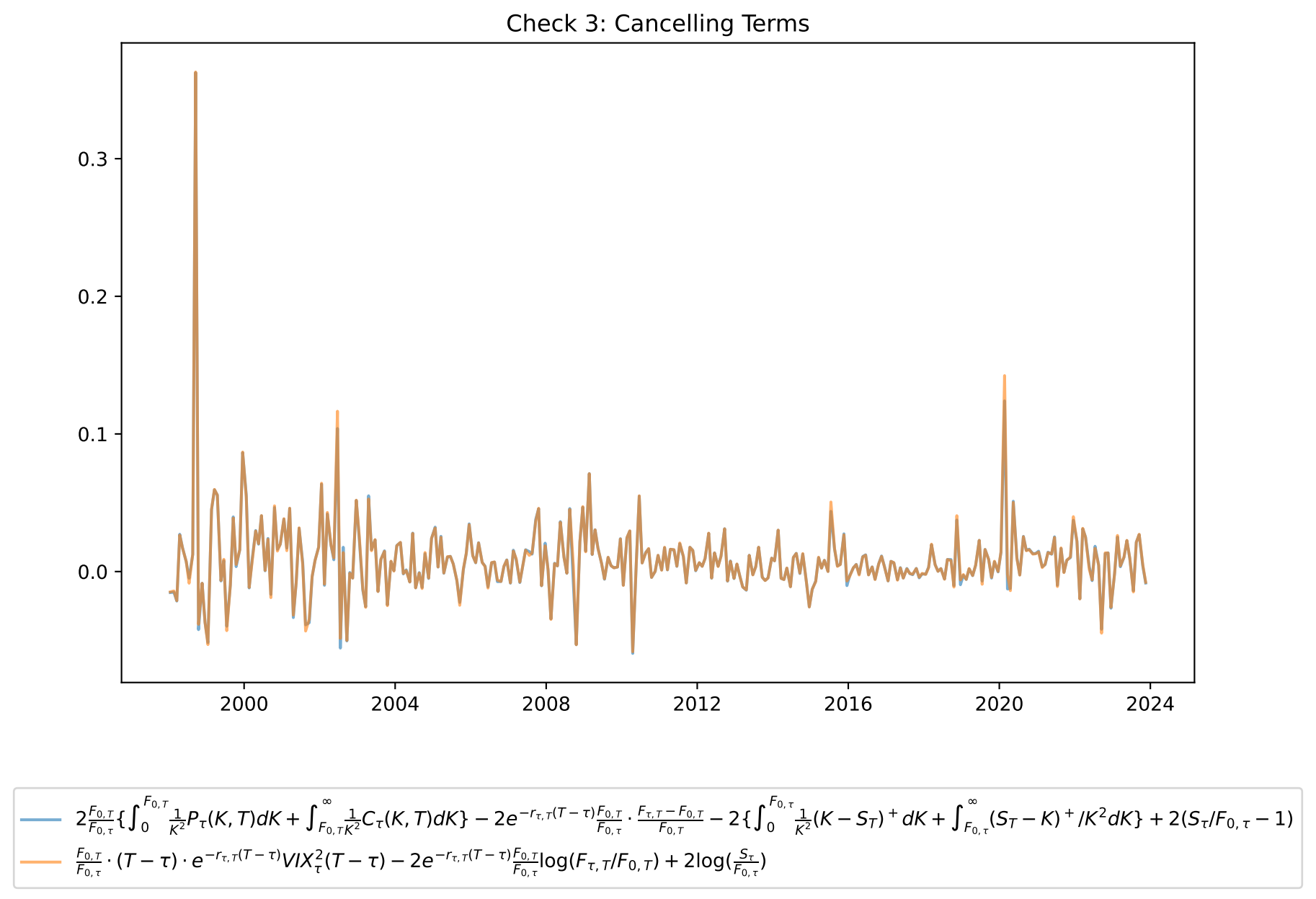
option_substitution_coverage
293.0
0.9999
0.0014
0.9762
1.0
1.0
1.0
1.0

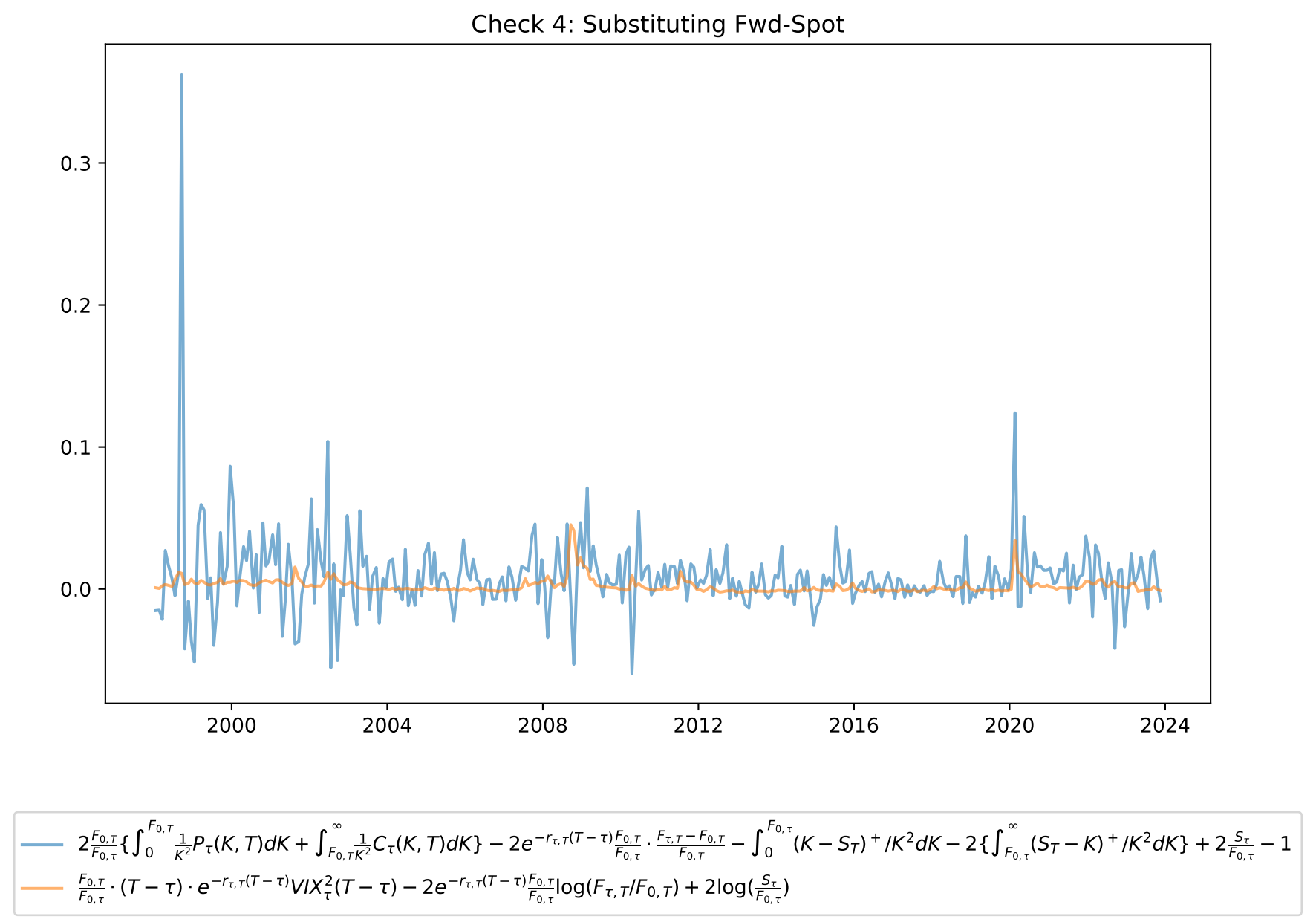

Ultimate (OLD)
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
2000
2004
2008
2012
2016
2020
2024
$2\{\int_0^{F_{0,T}} P_\tau(K,T)/K^2 dK + \int_{F_{0,T}}^{\infty} C_\tau(K,T)/K^2 dK\} + 2S_T/F_{0,\tau} - 1 - \int_0^{F_{0,\tau}} (K-S_T)^+/K^2 dK - \int_{F_{0,\tau}}^{\infty} (S_T-K)^+/K^2 dK$
$Eq.\,59 ending$

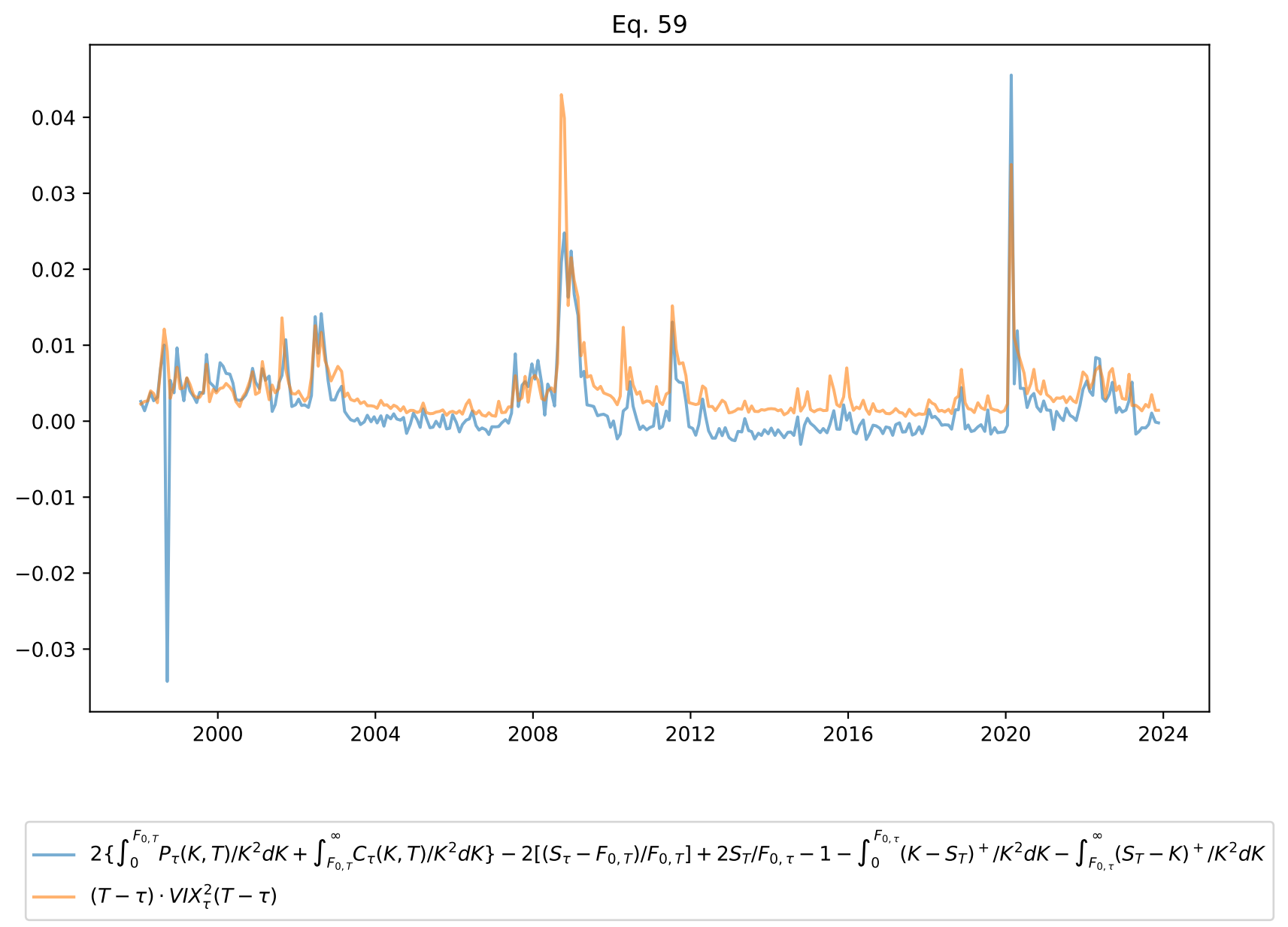
raw_portf_value	portf_aprrox_value_4
1.0	0.829195045519340 1
0.829195045519340 1	1.0


Check 1: log approximation
0.3
0.2
0.1
0.0
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-2\{\int_0^{F_{0,\tau}}\frac{1}{K^2}(K-S_T)^+dK+\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2(S_\tau/F_{0,\tau}-1)$
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}+2\log(S_\tau/F_{0,\tau})$

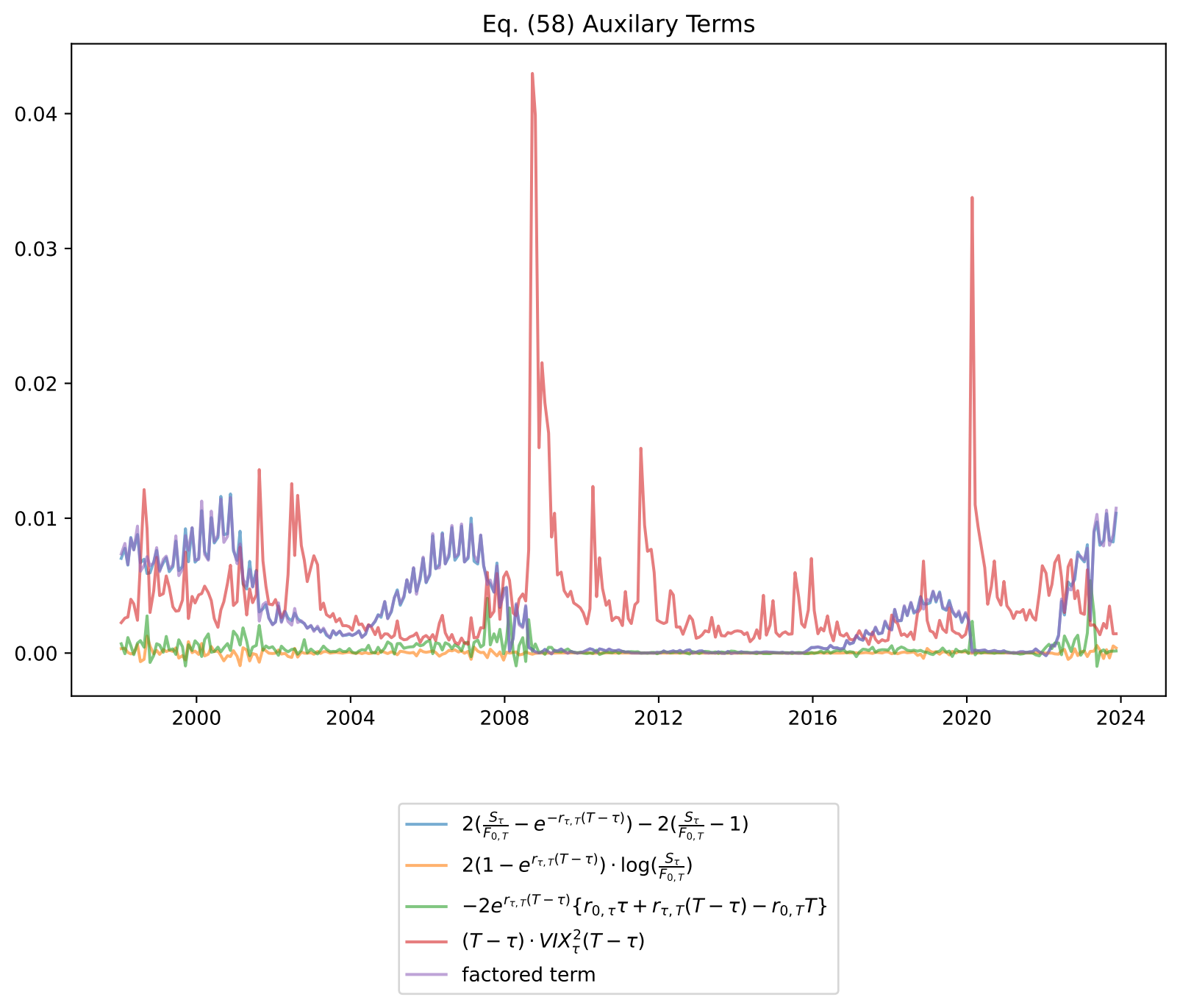

Check 2: Put-Call Parity Approximation
0.3
0.2
0.1
0.0
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-2\{\int_0^{F_{0,\tau}}\frac{1}{K^2}(K-S_T)^+dK+\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2(S_\tau/F_{0,\tau}-1)$
$\frac{F_{0,T}}{F_{0,\tau}}\cdot(T-\tau)\cdot e^{-r_{\tau,T}(T-\tau)}VIX_\tau^2(T-\tau)+2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}[\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-\log(F_{\tau,T}/F_{0,T})]-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}+2\log(\frac{S_\tau}{F_{0,\tau}})$


Check 3: Cancelling Terms
0.3
0.2
0.1
0.0
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-2\{\int_0^{F_{0,\tau}}\frac{1}{K^2}(K-S_T)^+dK+\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2(S_\tau/F_{0,\tau}-1)$
$\frac{F_{0,T}}{F_{0,\tau}}\cdot(T-\tau)\cdot e^{-r_{\tau,T}(T-\tau)}VIX_\tau^2(T-\tau)-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\log(F_{\tau,T}/F_{0,T})+2\log(\frac{S_\tau}{F_{0,\tau}})$


Check 4: Substituting Fwd-Spot
0.3
0.2
0.1
0.0
2000
2004
2008
2012
2016
2020
2024
$2\frac{F_{0,T}}{F_{0,\tau}}\{\int_0^{F_{0,T}}\frac{1}{K^2}P_\tau(K,T)dK+\int_{F_{0,T}}^{\infty}\frac{1}{K^2}C_\tau(K,T)dK\}-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\cdot\frac{F_{\tau,T}-F_{0,T}}{F_{0,T}}-\int_0^{F_{0,\tau}}(K-S_T)^+/K^2dK-2\{\int_{F_{0,\tau}}^{\infty}(S_T-K)^+/K^2dK\}+2\frac{S_\tau}{F_{0,\tau}}-1$
$\frac{F_{0,T}}{F_{0,\tau}}\cdot(T-\tau)\cdot e^{-r_{\tau,T}(T-\tau)}VIX_\tau^2(T-\tau)-2e^{-r_{\tau,T}(T-\tau)}\frac{F_{0,T}}{F_{0,\tau}}\log(F_{\tau,T}/F_{0,T})+2\log(\frac{S_\tau}{F_{0,\tau}})$


Eq. 59
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
2000
2004
2008
2012
2016
2020
2024
$2\{\int_0^{F_{0,T}} P_\tau(K,T)/K^2 dK + \int_{F_{0,T}}^{\infty} C_\tau(K,T)/K^2 dK\} - 2[(S_\tau - F_{0,T})/F_{0,T}] + 2S_T/F_{0,\tau} - 1 - \int_0^{F_{0,\tau}} (K - S_T)^+/K^2 dK - \int_{F_{0,\tau}}^{\infty} (S_T - K)^+/K^2 dK$
$(T-\tau)\cdot VIX_\tau^2(T-\tau)$

raw_portf_value	0
1.0	0.762288255885938
0.762288255885938	1.0


Eq. (58) Auxilary Terms
0.04
0.03
0.02
0.01
0.00
2000
2004
2008
2012
2016
2020
2024
$2(\frac{S_\tau}{F_{0,T}} - e^{-r_{\tau,T}(T-\tau)}) - 2(\frac{S_\tau}{F_{0,T}} - 1)$
$2(1 - e^{r_{\tau,T}(T-\tau)}) \cdot \log(\frac{S_\tau}{F_{0,T}})$
$-2e^{r_{\tau,T}(T-\tau)}\{r_{0,\tau}\tau + r_{\tau,T}(T-\tau) - r_{0,T}T\}$
$(T-\tau) \cdot VIX_\tau^2(T-\tau)$
factored term